LAB 2

Time allowed: 60 minutes

Data provided

- The image `coins.png`: built into MATLAB (load it directly with `imread('coins.png')`).

- The handwritten-digit dataset distributed with the textbook (`ocr_example.zip`, containing training and test images of the digits 0–9, loaded with the `load_data.m` function as in Tutorial 19.1).

- The video `video_bai3.mp4`: recorded with a fixed camera, 30 seconds long, 15 frames per second, frame size 640×360.

Task 1 — Counting and classifying coins (3 points)

You are given the image `coins.png`, which shows coins on a dark background. Write a program that:

1. Reads the image and converts it to a binary image using an automatic (Otsu) threshold.

2. Applies morphological processing to fill the holes inside the coins and remove small noise.

3. Labels the connected components and extracts the area (Area) of each coin.

4. Based on area, classifies the coins into two types: **small** and **large**.

5. Prints to the Command Window: the total number of coins, the number of small coins, and the number of large coins; and displays the original image with a number marked at the centroid of each coin.

Knowledge applied: Chapter 13 (morphology), Chapter 15 (segmentation), Chapter 18 (region feature extraction).

Task 2 — Recognizing handwritten digits 0–9 (3.5 points)

Using the digit dataset distributed with the textbook (the same dataset as in Tutorial 19.1, Chapter 19), write a program that:

1. Loads the training set and the test set using the `load_data.m` function.

2. For each digit image, extracts a feature vector of **at least 4 features** measured with `regionprops` (for example: area, eccentricity, Euler number, extent, solidity, area-to-convex-area ratio, ...).

3. Trains a classifier (minimum distance to class means, or k-nearest neighbors) on the training set.

4. Evaluates it on the test set: print the **overall accuracy** and the 10×10 **confusion matrix**.

5. States briefly, in a comment block at the top of the file: the features you selected, and your observation on how feature selection affects accuracy.

Note: test data must not be used in any way during training. Knowledge applied: Chapter 18 (feature extraction), Chapter 19 (visual pattern recognition).

Task 3 — Counting how many times an object appears in a video (3.5 points)

You are given the video `video_bai3.mp4`, recorded with a fixed camera. A **red ball** appears and disappears several times in the video; the frame also contains static objects of other colors. Write a program that:

1. Reads the video with `VideoReader` and processes it frame by frame.

2. Detects the red ball in each frame (for example by color segmentation, or by background subtraction combined with color/shape selection).

3. Builds a binary "present/absent" sequence for the ball over time, and from this sequence determines **how many times** the ball appears.

4. Prints to the Command Window: the number of appearances and the **time interval** (start second – end second) of each appearance, according to the video timeline.

5. Plots the presence/absence signal of the ball over time and saves the plot with your submission.

Knowledge applied: Chapter 20 (video fundamentals), Chapter 22 (digital video processing techniques: background subtraction, segmentation, and object tracking).

— End of exam —

Good luck!